JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

APPLIED PHYSICS DEGREE PATHWAYS

Note: This document is intended to serve as a scheduling aid for Applied Physics students. The schedule information provided is based on historical patterns and expected scheduling. Course offerings and modalities are subject to change due to instructor availability, student demand, and program priority. Students should check SIS for official course schedule information and the Johns Hopkins University Academic Catalog for official degree requirements.

APPLIED PHYSICS CORE COURSES & ELECTIVES | GENERAL M.S. DEGREE

(10) Courses: (4) Core Courses [at least (4) from the first six listed] + (6) Electives [including at least (3) 700- or 800-level]

CORE COURSES

615.641
Mathematical Methods for Physics & Engineering
Every Term • Asynchronous

615.653
Classical Mechanics
Every Term • Asynchronous

615.671
Principles of Optics
Every Term • Asynchronous

615.642
Electromagnetics
FA / SP • Asynchronous

615.654
Quantum Mechanics
FA / SP • Asynchronous

615.680
Materials Science
FA / SP • Asynchronous

615.651
Statistical Mechanics and Thermodynamics
Every Term • Asynchronous

615.665
Modern Physics
SP / SU • Asynchronous

675.613
Bold Science
Every Term • Asynchronous

615.661
Introduction to Planetary Science
SP / SU • Asynchronous

615.662
Introduction to Astrophysics
FA • Asynchronous

615.663
Physics of Lunar Exploration
In Development • Async.

615.664
Introduction to Exoplanets
In Development • Async.

615.666
Earth Systems & AI
In Development • Async.

615.682
Materials Characterization
In Development • Async.

615.683
Materials in Extreme Environments
In Development • Async.

615.685
Quantum Measurement and Sensing
In Development • Async.

615.715
Applied Atomic and Molecular Physics
Spring • Asynchronous

615.744
Physics of Space Systems I
SU / FA • Asynchronous

615.745
Physics of Space Systems II
SP • Asynchronous

615.747
Sensors and Sensor Systems
FA / SP • Asynchronous

615.748
Introduction to Relativity
SP • Asynchronous

615.749
Applied Nuclear Physics
In Development • Async.

615.751
Modern Optics
Fall • Asynchronous

615.753
Plasma Physics
SU / FA • Asynchronous

615.755
Space Physics
FA / SP • Asynchronous

615.757
Solid State Physics
FA • Asynchronous

615.760
Physics of Semiconductor Devices
SP • Asynchronous

615.761
Introduction to Oceanography
SP • Asynchronous

615.762
Applied Computational Electromagnetics
SP • Synchronous Online

615.763
Infrared Sensors: Physics and Applications
SP • Asynchronous

615.765
Chaos and Its Applications
SU / FA • Asynchronous

615.769
Physics of Remote Sensing
SP • Asynchronous

615.772
Cosmology
SU • Asynchronous

615.775
Physics of Climate
SP • Asynchronous

615.776
Quantum Algorithms: Near Term Applications
SU / SP • Asynchronous

615.778
Optical System Design and Modelling
Every Term • Asynchronous

615.780
Optical Detectors and Applications
FA • Asynchronous

615.781
Quantum Information Processing
SP • Asynchronous

716.782
Optics and MATLAB
SP • Asynchronous

615.783
Experimental Quantum Computing
In Development • Async.

615.784
Superconducting Devices: Physics and Applications
FA / SU • Asynchronous

615.786
Introduction to Open Quantum Systems
In Development • Async.

615.800
Applied Physics Project
Every Term • Asynchronous

615.802
Directed Studies in Applied Physics
Every Term • Asynchronous

MATERIALS & CONDENSED MATTER CONCENTRATION

(4) Applied Physics Core Courses + (6) MCM Electives, including at least (3) 700-level courses

CORE COURSES

615.641
Mathematical Methods for Physics & Engineering
Every Term • Asynchronous

615.642
Electromagnetics
FA / SP • Asynchronous

615.651
Statistical Mechanics and Thermodynamics
Every Term • Asynchronous

615.680
Materials Science
FA / SP • Asynchronous

Electives

615.654
Quantum Mechanics
FA / SP • Asynchronous

615.682
Materials Characterization
In Development • Async.

615.683
Materials in Extreme Environments
In Development • Async.

515.617
Nanomaterials

515.635
Mechanical Properties of Materials

525.606
Electronic Materials

525.621
Introduction to Electronics & the Solid State

535.684
Modern Polymeric Materials

700-level courses – Select at least 3

615.715
Applied Atomic and Molecular Physics
Spring • Asynchronous

615.747
Sensors and Sensor Systems
FA / SP • Asynchronous

615.757
Solid State Physics
FA • Asynchronous

615.760
Physics of Semiconductor Devices
SP • Asynchronous

615.763
Infrared Sensors: Physics and Applications
SP • Asynchronous

615.784
Superconducting Devices: Physics and Applications
FA / SU • Asynchronous

615.800
Applied Physics Project
Every Term • Asynchronous

615.802
Directed Studies in Applied Physics
Every Term • Asynchronous

535.732
Fatigue and Fracture of Materials

535.748
Stress Waves, Impacts, and Shockwaves

Unshaded boxes indicate elective courses not offered by the Applied Physics program. Check the EP website / SIS for course offering information.

APPLIED OPTICS & PHOTONICS CONCENTRATION

(4) Applied Physics Core Courses + (1) ECE Core Course + (3) 700-level Electives + (2) Additional Electives

CORE COURSES
Applied Physics

Select (4) of the following

615.641 * Mathematical Methods for Physics & Engineering
Every Term • Asynchronous

615. 642 Electromagnetics
FA / SP • Asynchronous

615.653 Classical Mechanics
Every Term • Asynchronous

615.654 Quantum Mechanics
FA / SP • Asynchronous

615.665 Modern Physics
SP / SU • Asynchronous

615.671 Principles of Optics
Every Term • Asynchronous

Unshaded boxes indicate elective courses not offered by the Applied Physics program. Check the EP website / SIS for course offering information.

CORE COURSES
Electrical & Computer Engineering

Select (1) of the following

525.613 Fourier Techniques in Optics

525.625 Laser Fundamentals

525.691 Fundamentals of Photonics

* Students interested in taking 615.653/615.654 must complete 615.641 as a prerequisite

ELECTIVES

525.613 Fourier Techniques in Optics

525.625 Laser Fundamentals

525.636 Optics & Photonics Lab

525.674 Image Processing and Analysis

525.691 Fundamentals of Photonics

Select at least (3) of the following 700-level courses

615.751 Modern Optics
Fall • Asynchronous

615.778 Optical System Design and Modelling
Every Term • Asynchronous

615.780 Optical Detectors and Applications
FA • Asynchronous

615.781 Quantum Information Processing
SP • Asynchronous

716.782 Optics and MATLAB
SP • Asynchronous

525.753 Laser Systems and Applications

525.756 Optical Propagation, Sensing, and Backgrounds

525.772 Fiber Optic Communication Systems

525.797 Advanced Fiber Optic Laboratory

525.796 Introduction to High-Speed Optoelectrics

585.734 Biophotonics

SPACE AND EARTH SCIENCE FOCUS AREA

(4) Space Science Core Courses + (2) Earth Science, (2) Heliophysics & Planetary Science, (2) Astrophysics Electives

CORE COURSES

Select 4 of the following, including 675.613, followed by 615.744

Required

- 675.613 Bold Science — Every Term • Asynchronous
- 675.744 Physics of Space Systems I — SU / FA • Asynchronous
- 615.641 Mathematical Methods for Physics & Engineering — Every Term • Asynchronous
- 615.642 Electromagnetics — FA / SP • Asynchronous
- 615.653 Classical Mechanics — Every Term • Asynchronous
- 615.654 Quantum Mechanics — FA / SP • Asynchronous
- 615.665 Modern Physics — SP / SU • Asynchronous
- 615.671 Principles of Optics — Every Term • Asynchronous
- 615.651 Statistical Mechanics and Thermodynamics — Every Term • Asynchronous
- 615.680 Materials Science — FA / SP • Asynchronous

EARTH SCIENCE ELECTIVES

Select 2 of the following

- 615.666 Earth Systems & AI — In Development • Async.
- 615.761 Introduction to Oceanography — SP • Asynchronous
- 615.769 Physics of Remote Sensing — SP • Asynchronous
- 615.775 Physics of Climate — SP • Asynchronous

HELIOPHYSICS AND PLANETARY SCIENCE ELECTIVES

Select 2 of the following

- 615.661 Introduction to Planetary Science — SP / SU • Asynchronous
- 615.663 Physics of Lunar Exploration — In Development • Async.
- 615.664 Introduction to Exoplanets — In Development • Async.
- 615.745 Physics of Space Systems II — SP • Asynchronous
- 615.755 Space Physics — FA / SP • Asynchronous

ASTROPHYSICS ELECTIVES

Select 2 of the following

- 615.662 Introduction to Astrophysics — FA • Asynchronous
- 615.748 Introduction to Relativity — SP • Asynchronous
- 615.772 Cosmology — SU • Asynchronous

HARDWARE ELECTIVES

OPTIONAL: Select up to 2 of the following to substitute other electives

- 615.747 Sensors and Sensor Systems — FA / SP • Asynchronous
- 615.763 Infrared Sensors: Physics and Applications — SP • Asynchronous



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